Objective

This assignment aims to explore passage retrieval techniques using Natural Language Processing (NLP). You will do so using two methods: lexical retrieval with TF-IDF and semantic search with miniLM. After implementing both approaches, you'll evaluate their effectiveness and propose improvements.

As a DS in Sleek, your work will always revolve around cyber-related data. However, to ease this assignment, your retrieval efforts will be performed on the text of the first Harry Potter book.

Tasks

Download the Text

- [ ] Download the text of the 1st Harry Potter book.

Lexical Retrieval

- [ ] Demonstrate retrieving the top 5 passages per example search using TF-IDF scoring. Repeat this step to demonstrate three different examples (resulting in 15 passages in total).
- [ ] Discuss the main issues with the TF-IDF approach.

Semantic Search

- [ ] Load the miniLM model from HuggingFace. Note that there are many versions of this model, so be sure to explain the reasoning for your choice.
- [ ] Divide the Harry Potter book into chunks and load the data into Faiss for efficient similarity search.
- [ ] Evaluate the generated vector space by checking uniformity and alignment.
- [ ] Similar to the requirement in the lexical search, showcase the retrieval of the top 5 passages retrieved for each of the 3 example searches.
- [ ] *Bonus:* Implement an interactive retrieval system allowing users to retrieve the top 5 passages per a search line of their liking.

Evaluation

- [ ] Utilize the results obtained from miniLM as ground truth. Evaluate the TF-IDF results against them using the P@k metric and two more metrics of your liking.

Open Qs

- [ ] Discuss potential improvements for retrieval using an LLM. Clarify whether this approach applies to lexical, semantic, or both searches.
- [ ] Highlight the drawbacks or limitations of your suggestion.

- [ ] Explain how you would create a reliable ground truth for evaluation (instead of simply using the miniLM results). How would you tag each chunk?

Deliverables

- [ ] Implementation of TF-IDF and semantic search with miniLM.
- [ ] Demonstrated retrieval examples for both methods.
- [ ] Evaluation of TF-IDF results against miniLM ground truth.
- [ ] *Bonus:* Interactive retrieval system implementation.
- [ ] Detailed discussion on improvements with an LLM and its limitations.
- [ ] Discussion on establishing a ground truth dataset for evaluation.

Submission Guidelines

Please ensure your completed assignment is submitted as a GitHub repository, including clear documentation and instructions. You should dedicate up to 6 hrs to the exercise, spread out over a maximum of three days.

Note

A partial solution, showing off your way of thinking and capabilities, is better than no solution at all. Feel free to reach out for any clarification or assistance during the assignment.

Good luck!